

Sat

4/25/2026

Fuzzy Hw

19)

$$f(x) = \max \left\{ \alpha \min \left(\frac{x-a}{b-a}, 1, \frac{d-x}{d-c} \right), 0 \right\}$$

$$a < b < c < d, \quad 0 < \alpha \leq 1$$

gives trap set determined by $(a, 0, b, \alpha), (c, \alpha, d, 0)$

Proof:

$$\text{Let } u(x) = \frac{x-a}{b-a}, \quad v(x) = \frac{d-x}{d-c}, \quad m(x) = \min(u, 1, v)$$

$$f(x) = \max \{ \alpha m(x) \} = \max \{ \alpha \min(u, 1, v) \}$$

$$(i) \quad x \leq a, \quad u(x) = \frac{x-a}{b-a} \leq 0, \quad f(x) = \max \{ \alpha \min(\leq 0, 1, v) \} = \max \{ \alpha \leq 0 \} = 0. \quad \text{Outer max } f(x) = 0$$

$$(ii) \quad a \leq x \leq b, \quad u = \frac{x-a}{b-a} \in [0, 1], \quad 1 \text{ constant}$$

$$b < c < d, \quad d-x \geq d-b > d-c \rightarrow v = \frac{d-x}{d-c} \geq \frac{d-b}{d-c} \geq 1$$

$$\text{thus, } m = u \rightarrow f(x) = \max \left\{ \alpha \frac{x-a}{b-a}, 0 \right\} = \alpha \frac{x-a}{b-a}$$

Linear rise from $(a, 0)$ to (b, α)

$$(iii) \quad b \leq x \leq c, \quad u = \frac{x-a}{b-a} \geq 1, \quad v = \frac{d-x}{d-c} \geq 1$$

So $m = 1, f(x) = \alpha$. Flat top from (b, α) to (c, α)

$$(iv) \quad c \leq x \leq d, \quad v = \frac{d-x}{d-c} \in [0, 1], \quad u = \frac{x-a}{b-a} > 1$$

$$m = v, \quad f(x) = \alpha \frac{d-x}{d-c}. \quad \text{So linear decrease } (c, \alpha) \text{ to } (d, 0)$$

$$(v) \quad x \geq d, \quad v = \frac{d-x}{d-c} \leq 0, \quad m \leq 0, \quad f(x) = 0$$

Fuzzy MH

Thus the five cases describe the straight-line segments connecting $(a, 0) \rightarrow (b, a) \rightarrow (c, a) \rightarrow (d, 0)$ the trapezoidal set

$$20) \quad O_{(5,5)}(z) = (A_1(5) \wedge B_1(5) \wedge C_1(5)) \vee (A_2(5) \wedge B_2(5) \wedge C_2(5))$$

$A[a, b, c]$ is triangular set determined by the points $(a, 0), (b, 1), (c, 0)$

membership values at input point $(x_1, x_2) = (5, 5)$

Set	triangle peak	value at 5	$A(x) = \begin{cases} 0 & x \leq a & (1) \\ \frac{x-a}{b-a} & a \leq x \leq b & (2) \\ \frac{c-x}{c-b} & b \leq x \leq c & (3) \\ 0 & x \geq c & (4) \end{cases}$
(3) $A_1[a, b, c]$	4	$\frac{(5)-4}{6-4} = \frac{1}{2} = .5$	
(3) $A_2[4, 6, 8]$	6	$\frac{(5)-4}{6-4} = \frac{1}{2} = .5$	
(2) $B_1[1, 4, 6]$	4	$\frac{6-5}{6-4} = \frac{1}{2} = .5$	
(2) $B_2[4, 8, 9]$	8	$\frac{8-5}{8-4} = \frac{3}{4} = .75$	

$$\text{Mamdani: } w_x = A_x(x) \wedge B_x(x) = \min(A(x), B(x))$$

rule firing strength $w_1 = A_1 \wedge B_1(5) = \min(.5, .5) = .5$

rule firing strength $w_2 = A_2(5) \wedge B_2(5) = \min(.5, .75) = .5$

rule firing strength C_1 is capped at $w_1 = .5$

rule firing strength $C_1[1, 4, 7]$ capped at $w_1 = 0.5$ (sub x w / z)

rising edge: $\frac{(z-1)}{b-a} = \frac{z-1}{4-1} = \frac{z-1}{3} = .5 \Rightarrow z-1 = 1.5, z = 2.5$

falling edge: $\frac{(c-z)}{c-b} = \frac{7-z}{7-4} = \frac{7-z}{3} = .5 \Rightarrow 7-z = 1.5, z = 5.5$

$C_2[3, 6, 9]$ capped at $w_2 = .25$

rising edge $\frac{(z-3)}{b-a} = \frac{z-3}{6-3} = \frac{z-3}{3} = .25 \Rightarrow z-3 = .75, z = 3.75$

falling edge $\frac{c-z}{c-b} = \frac{9-z}{9-6} = \frac{9-z}{3} = .25 \Rightarrow 9-z = .75, z = 8.25$
 $[3.75, 8.25]$

Fuzzy HW Notes

Fuzzification: Evaluate each antecedent MF of the input. With $(x_1, x_2) = (5, 5) \rightarrow A_1(\) A_2 \dots$ values are called membership degrees

Problem 20

Rule firing strength

R_1 : IF x_1 is A_1 AND x_2 is B_1 then z is C_1

R_2 : IF x_1 is A_2 AND x_2 is B_2 then z is C_2

$(x_1, x_2) = (5, 5)$ want to know

Gap 3

$$w_i = A_i(5) \wedge B_i(5) = \min(A_i(5), B_i(5))$$

Gap 4

rule i's output = $\min(w_i, C_i(z))$

~~remove~~ remove everything above w_i

Gap 5

OR is S norm $\rightarrow$ max (upper envelope)

Gap 6 Building Piecewise Table

20 continued)

$$C_{\text{fb}}(z) = \max(\min(0.5, C_1(z)), \min(0.25, C_2(z)))$$

rule 1 trapezoid $z = 1, 2.5, 5.5, 7$

rule 2 trapezoid $z = 3, 3.75, 6.25, 9$

Crossover:

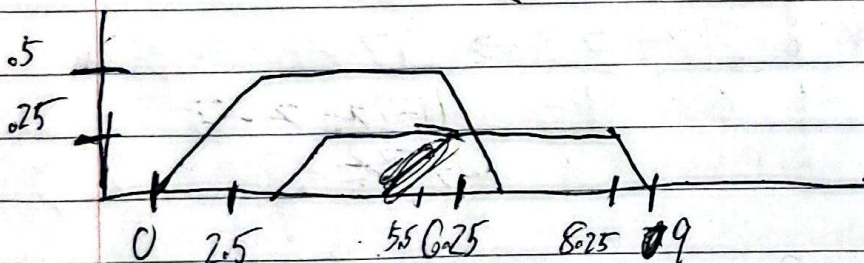
$$\frac{7-z}{3} = 0.25 \rightarrow \text{crossover} \quad 7-z = 0.75, z = 6.25$$

Fuzzy HW

z range	$G(5,5)(z)$
$z < 1$	0
$[1, 2.5]$	$\frac{z-1}{3}$
$[2.5, 5.5]$	0.5
$[5.5, 6.25]$	$\frac{z-5.5}{3}$
$[6.25, 8.25]$	0.25
$[8.25, 9]$	$\frac{9-z}{3}$
$z > 9$	0

MOM: smallest & largest z at which $O(z)$ has global max
highest 0.5 at $[2.5, 5.5]$

$$z^*_{MOM} = \frac{z_{min} + z_{max}}{2} = \frac{2.5 + 5.5}{2} = 4.0$$



2.1)

Larsen agg & FOM

$$w_1 = \min(A_1(5), B_1(5)) = 0.5, w_2 = \min(A_2(5), B_2(5)) = 0.25$$

$$\text{Larsen output} = w_i \cdot G_i(z)$$

Rule 1: 0.5 $G_1(z)$ $[1, 4, 7]$ peak $z=4$

new peak $0.5 \cdot 1 = 0.5$ $z=4$

$$\frac{1}{z} \frac{(z-1)}{3} = \frac{z-1}{6} \quad [1, 4]; \quad \frac{1}{z} \frac{(7-z)}{3} = \frac{7-z}{6} \quad [4, 7]$$

Rule 2:

$C_2 [3, 6, 9]$, peak 1 at $z=6$
 peak of $.25 \cdot 1 = .25$, $z=6$; support $[3, 9]$

$$\frac{z-3}{12}, [3, 6]; \quad \frac{9-z}{12}, [6, 9];$$

$$0.5 C_{11}(z) = \begin{cases} (z-1)/6 & z \in [1, 4] \\ (7-z)/6 & z \in [4, 7] \\ 0 & \text{else} \end{cases} \quad 0.25 C_2(z) = \begin{cases} (z-3)/12 & z \in [3, 6] \\ (9-z)/12 & z \in [6, 9] \\ 0 & \text{else} \end{cases}$$

Crossovers

$C_1 [1, 4]$ & $C_2 [3, 6] : \frac{z-1}{6} = \frac{z-3}{12} \rightarrow 2(z-1) = z-3$

$2z-2 = z-3$
 $z = -1 \quad \times$

$[4, 7] : \frac{7-z}{6} = \frac{z-3}{12} \rightarrow 2(7-z) = z-3$

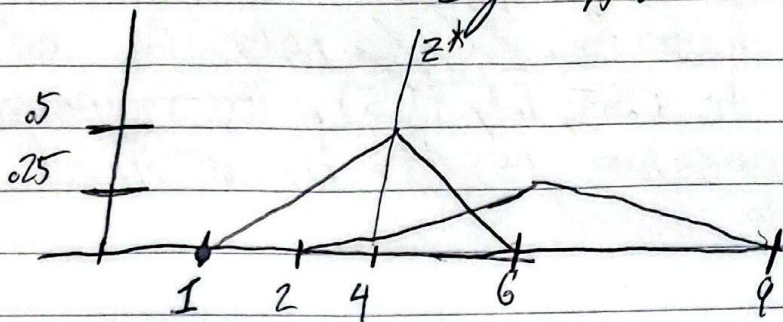
$14-2z = z-3$
 $-3z = -17, z = \frac{17}{3};$

$[6, 9] : \frac{9-z}{12} = \frac{7-z}{6} \quad 14-2z = 9-z, z = 5 \quad \times$
 max

$$C_{(5,5)}(z) = \begin{cases} 0 & z < 1 \\ \frac{z-1}{6} & 1 \leq z \leq 4 \\ \frac{7-z}{6} & 4 \leq z \leq \frac{17}{3} \\ \frac{z-3}{12} & \frac{17}{3} \leq z \leq 6 \\ \frac{9-z}{12} & 6 \leq z \leq 9 \\ 0 & z > 9 \end{cases}$$

$z^*_{FOV} = 4.0$

Fuzzy HW



7)

(5/55) $A_1 = A_2 = R_1 = R_2 = 1$
 $\dot{x} = \begin{bmatrix} -1 & 1 \\ 1 & -2 \end{bmatrix} x + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u, \quad y = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} x$

$x_1 - 2x_2 = 0$

$h_{10} = 1m, \dot{x} = 0, x_1 - 2x_2 = 0 \Rightarrow h_2 = h_1 / 2 = 0.5m$

$-h_1 + h_2 + u_{ss} = 0 \Rightarrow u_{ss} = h_1 - h_2 = 0.5m/s$

(C)

The fuzzy controller maintains tank levels at set points. h_1 settles at 1.005m.

h_2 at 0.502m. 0.5% overshoot

PID is faster (3 seconds vs 6.4+1.2s for rise & settling time), but fuzzy controller smoother & has steady state accuracy

To tune controller had to adjust 3 external gains
 G_e - error

G_{de} - rate of change of error

G_u - fuzzy output before the integrator

first attempt at (1.5, 0.6, 0.8): 35% overshoot

had to lower G_e , ~~the~~ raise G_d and
 lower G_v to (0.85, 1.1, 0.5). Too aggressive
 Too aggressive results in overshoot
 Too gentle caused offset.

Only inflow to tank 1 is controllable, so h_2 is
 rigidly tied to h_1 through coupling resistance &
 always settles at half the value of h_1 . Controlling
 h_2 independently needs 2nd actuator

	$\dot{e} = NE$	$e = ZE$	$\ddot{e} = PE$
$e = NE$	NV	NV	ZV
$e = ZE$	NV	ZV	PV
$e = PE$	ZV	PV	PV

PID Gains Chosen:

$K_p = 4.0$ $K_i = 1.2$, $K_d = 0.8$, derivative filter $N=2$

Fuzzy Gains

$G_e, G_d, G_v = 0.85, 1.1, 0.5$;